Belief-State Tracking:

A Comparative Study of DQN and DRQN on a Two-Goal Memory POMDP

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***Abstract*—We investigate the fundamental limitation of memoryless Deep Q-Networks (DQN) in Partially Observable Markov Decision Processes (POMDPs) and demonstrate how Deep Recurrent Q-Networks (DRQN) overcome this limitation through temporal belief-state encoding. We construct a purpose-built 11×11 two-goal grid-world POMDP in which a one-time hint at episode initialization identifies the correct navigation goal; all subsequent observations are informationally identical regardless of the active goal, establishing a provable theoretical ceiling of approximately 50% correct-goal success for any memoryless policy. Standard DQN converges near this ceiling at 41.0% correct-goal success and a mean reward of −1.91, confirming the theoretical prediction. DRQN, augmented with an auxiliary hint-prediction head that provides a direct one-step gradient path to the hint encoder, achieves 92.5% correct-goal selection and a mean reward of +6.21—a 51.5 percentage-point improvement over DQN. PCA visualization of the DRQN's LSTM hidden states reveals two cleanly separated clusters corresponding to the two goal identities, with PC1 explaining 76.1% of total hidden-state variance, constituting direct mechanistic evidence of learned belief-state formation. Our environment, auxiliary training technique, and benchmark methodology provide a reproducible framework for studying memory requirements in recurrent reinforcement learning.**

***Index Terms*—Deep Q-Network (DQN), Deep Recurrent Q-Network (DRQN), Partially Observable Markov Decision Process (POMDP), Belief-State Tracking, Long Short-Term Memory (LSTM), Auxiliary Tasks, Memory in Reinforcement Learning**

I. INTRODUCTION

Reinforcement learning (RL) has demonstrated remarkable success across fully observable domains, from board games to robotic control. Real-world decision-making, however, is inherently partial: critical state information is often hidden, delayed, or available for only a limited window of time. Such settings are formally characterized as Partially Observable Markov Decision Processes (POMDPs), and they expose a fundamental limitation of standard memoryless RL algorithms.

Deep Q-Networks (DQN) [1] approximate the action-value function using a deep neural network conditioned on the current observation. While this approach achieved landmark results on Atari games, it remains inherently memoryless: any information absent from the current observation is irrecoverably lost. In tasks where the optimal action depends on past observations, DQN is provably suboptimal regardless of network capacity, training duration, or hyperparameter configuration.

This work constructs a minimal, precisely controlled POMDP—a two-goal memory grid-world—that isolates this limitation with mathematical precision. A one-time hint at episode initialization reveals which of two goals is active; thereafter, all observations are informationally identical for both goal types. This design imposes a hard theoretical ceiling of ≈50% correct-goal success on any memoryless policy, making DQN's failure analytically predictable and empirically verifiable.

We compare DQN against Deep Recurrent Q-Networks (DRQN) [2], which incorporate an LSTM layer capable of integrating temporal context. Crucially, we augment DRQN with an auxiliary hint-prediction head—a task-specific objective that creates a direct gradient path to the hint encoder at step 0, substantially improving training efficiency and final performance. PCA of the trained DRQN's LSTM hidden states provides mechanistic evidence of genuine belief-state formation.

Our principal contributions are: a POMDP benchmark with analytically derived performance bounds for memoryless agents; an auxiliary hint-prediction head for DRQN shown to be essential for reliable convergence; and PCA-based mechanistic evidence of belief-state formation in a recurrent RL agent.

II. RELATED WORK

A. Deep Q-Networks

Mnih et al. [1] introduced DQN, combining Q-learning with deep neural network function approximation, experience replay, and a frozen target network. Despite achieving human-level performance on over 40 Atari games, DQN's single-observation input fundamentally limits applicability to POMDPs, as any information absent from the current time step is irrecoverably lost to the policy.

B. Deep Recurrent Q-Networks

Hausknecht and Stone [2] proposed DRQN, replacing DQN's first fully connected layer with an LSTM. Unrolled over a sequence of observations, the LSTM maintains a compressed, implicitly learned belief state over the observation history. The authors demonstrated improved performance under frame-flicker partial observability and noted that DRQN's advantage becomes decisive precisely when observability reduction is substantial—exactly the condition our environment enforces.

C. Auxiliary Tasks in Recurrent RL

Jaderberg et al. [3] demonstrated that auxiliary prediction objectives—predicting pixel changes, reward signals, or environment depth—dramatically accelerate representation learning and improve sample efficiency in deep RL. More recent systems including Dreamer [4] and R2D2 [5] incorporate auxiliary objectives into their recurrent architectures. Our hint-prediction head follows this tradition, providing a task-specific signal that directly incentivizes the encoder to treat the transient one-time hint as a high-salience input.

D. Memory Benchmarks in Reinforcement Learning

The T-maze [6] is a classical POMDP benchmark requiring long-range memory of a single directional cue. Bakker [7] first applied recurrent RL to T-maze variants. The POPGym benchmark suite [8] provides standardized partially observable environments. Our two-goal memory POMDP extends this tradition with a theoretically grounded design that decouples the memory requirement from navigation difficulty via the dual-compass observation structure, ensuring both agents have identical navigational knowledge.

III. PROBLEM FORMULATION

A. POMDP Framework

A POMDP is defined by the tuple $(S, A, O, T, \Omega, R, \gamma)$, where S is the state space, A the action space, O the observation space, $T(s'|s, a)$ the state transition function, $\Omega(o|s', a)$ the observation function, $R(s, a)$ the reward function, and $\gamma \in [0, 1)$ the discount factor. Unlike in an MDP, the agent receives observation $o_t \in O$ rather than the true state $s_t \in S$. Optimal behavior requires conditioning on the full observation history $h_t = (o_0, a_0, o_1, a_1, \ldots, o_t)$.

A belief state $b(s) = P(s_t \mid h_t)$ summarizes the agent's uncertainty over the true state given the observation history. Computing the exact belief state is intractable at scale; recurrent networks provide a practical mechanism for learning a compressed approximation implicitly through the hidden-state dynamics.

B. The Two-Goal Memory Environment

We implement an 11×11 discrete grid-world with two fixed goals at opposite corners: goal_A = (1, 9) and goal_B = (9, 1). At episode start, one goal is uniformly randomly designated as the active (correct) target. The 14-dimensional observation vector comprises:

- Dims 0–8: 3×3 binary field-of-view (FOV) centered on the agent, with mild peripheral sensor noise ($p = 0.1$ flip probability). The centre cell is noiseless.
- Dim 9: One-time hint signal $h = +1$ if goal_A is active, $h = -1$ if goal_B is active, and $h = 0$ for all steps $t > 0$. This is the sole goal-identity signal.
- Dims 10–13: Dual noise-free compass directions toward both goals (row-sign, col-sign for each).

The dual compass structure is critical for experimental rigour: it ensures both DQN and DRQN always possess perfect directional knowledge toward both goals. The only thing distinguishing a correct from incorrect action is goal identity, which is observable only at $t = 0$. Environment parameters are summarized in Table I.

TABLE I
ENVIRONMENT CONFIGURATION

Parameter	Value
Grid size	11 × 11
Number of obstacles	15
goal_A position	(row 1, col 9)
goal_B position	(row 9, col 1)
Observation dim.	14
Actions	4 (Up/Down/Left/Right)
Max episode steps	150
Correct goal reward	+10.0
Wrong goal penalty	−5.0
Step penalty	−0.1 / step
Peripheral sensor noise	p = 0.1 (FOV)

C. Theoretical Analysis of DQN Failure

For $t > 0$, DQN's input carries hint = 0. Both goal-A and goal-B episodes produce observationally identical inputs at all post-hint steps. The replay buffer therefore contains identical observation vectors paired with contradictory Q-targets:

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Goal-A episode: obs_t → optimal = follow
compass_A
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Gradient descent averages these contradictory targets, converging to a policy that follows neither compass reliably. The best possible memoryless strategy—committing unconditionally to one goal—yields:E[R | memoryless] = 0.5 · (+10) + 0.5 · (−5) − 0.1 · s = 2.5 − 0.1s

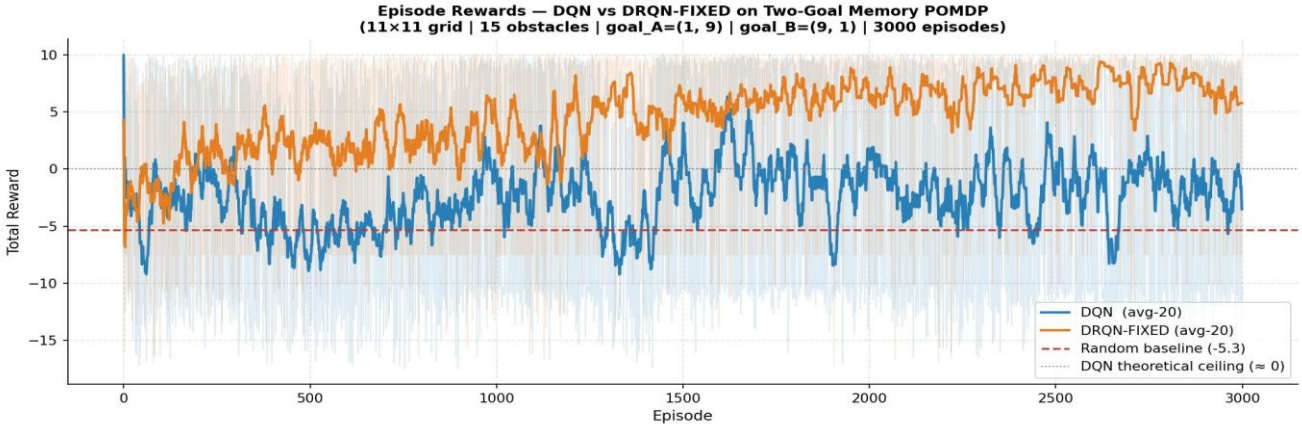


Fig. 1. Training reward curves (20-episode moving average) for DQN (blue) and DRQN (orange) over 3,000 training episodes on the Two-Goal Memory POMDP.

IV. METHODOLOGY

A. DQN Architecture and Training

The DQN model comprises three fully connected layers: Linear(14→256, ReLU) → Linear(256→256, ReLU) → Linear(256→4). It is conditioned on a single 14-dimensional observation with no temporal context. Training uses the Adam optimizer (lr = 1×10^{-3}), discount factor $\gamma = 0.99$, and ε-greedy exploration annealing from $\varepsilon = 1.0$ to $\varepsilon = 0.05$ at a decay rate of 0.995 per episode. A transition-level experience replay buffer (capacity 10,000) is sampled uniformly in mini-batches of 64.

B. DRQN Architecture and the Auxiliary Hint Head

The DRQN model comprises four components: a linear encoder Linear(14→128, ReLU), a single-layer LSTM(128→128) processing the encoded sequence and maintaining the temporal belief state, a Q-head Linear(128→4), and an auxiliary hint head Linear(128→1) trained with binary cross-entropy loss to predict which goal is active (target: 1.0 = goal_A, 0.0 = goal_B) from the LSTM output at each timestep.

The auxiliary hint head is the single most consequential architectural decision in this implementation. Without it, the encoder at step 0 receives gradients from Q-learning only via truncated backpropagation through time (TBPTT) across the full burn-in sequence—a long-range, weak gradient path. The auxiliary head creates a 1-step gradient path: loss at t=0 → hint head → LSTM output → encoder → obs[9]. This auxiliary prediction technique is well-established in modern recurrent RL systems [3, 4, 5].

C. Sequence-Level Replay Buffer for DRQN

Standard transition-level replay destroys temporal context for recurrent agents. The DRQN replay buffer stores complete episodes; during training, contiguous sub-sequences of length seq_len = 20 are sampled uniformly. The first burn_in = 10 steps are used exclusively to warm up the LSTM hidden state; Q-learning targets are computed and backpropagated only on the remaining (seq_len − burn_in) = 10 steps, preventing gradient corruption from uninitialized hidden states.

D. Architecture Summary

Table II summarizes the architectural differences between DQN and DRQN. Both agents were trained for 3,000 episodes on the same fixed-seed environment instance (seed = 42), ensuring identical navigation challenge layouts. Evaluation was conducted over 200 independent episodes with $\varepsilon = 0$ (fully greedy) and fresh zero-initialized LSTM hidden states for DRQN.

TABLE II
ARCHITECTURE COMPARISON: DQN VS. DRQN

Component	DQN	DRQN
Input encoder	Linear(14→256, ReLU)	Linear(14→128, ReLU)
Memory module	None	LSTM(128→128)
Q-head	Linear(256→4)	Linear(128→4)
Auxiliary head	—	Linear(128→1)
Aux. loss fn.	—	BCE (goal pred.)

V. Mathematical Framework

The DQN loss uses temporal-difference learning with a frozen target network (θ^-):

$$L(\theta) = E[(r + \gamma \max_{a'} Q(s',a';\theta^-) - Q(s,a;\theta))^2]$$

DRQN replaces single-step conditioning with recurrent hidden-state updates. At each step t, the LSTM propagates the belief state:

$$h_t, c_t = LSTM(f\varphi(o_t), h_{t-1}, c_{t-1}); \quad Q(o_t, a; \theta) = W_q \cdot h_t$$

The auxiliary hint-prediction head appends a Binary Cross-Entropy loss at step 0, creating a direct one-step gradient path to the encoder:

$$L_hint = -[y \log \sigma(\hat{y}) + (1-y) \log(1-\sigma(\hat{y}))]$$

where $y \in \{0,1\}$ is the true goal identity, $\hat{y} = W_h \cdot h_0$ is the hint-head logit from LSTM output at step 0, and σ is the sigmoid. The combined training objective is:

$$L_total = L_TD + \lambda \cdot L_hint$$

The expected reward for DRQN with correct-goal accuracy p and step penalty $\delta = 0.1\bar{1}s$ is:

$$E[R_DRQN] = p(+10-\delta) + (1-p)(-5-\delta) = 15p - 5 - \delta$$

VI. EXPERIMENTAL RESULTS

A. Training Dynamics

Fig. 1 shows the training reward curves for both agents over 3,000 episodes, smoothed with a 20-episode rolling average. DRQN exhibits a consistent upward learning trajectory, surpassing the DQN theoretical ceiling (reward ≈ 0) by episode ≈ 200 and reaching a stable plateau near +5 to +7 reward by episode 1,500. DQN oscillates between −1 and −4, never crossing the theoretical ceiling throughout training.

The rapid initial improvement in DRQN reward reflects the effectiveness of the auxiliary hint-prediction head in establishing goal-identity encoding early in training. The DRQN standalone training curve (Fig. 5, provided in supplementary material) confirms that the upward trend is robust across the full 3,000-episode training run. The random baseline at −5.35 mean reward is shown for reference; both trained agents substantially exceed this threshold.

B. Evaluation Performance

Fig. 2 presents quantitative evaluation results across 200 greedy episodes. DRQN achieves a mean reward of +6.21 (last 100 training episodes) and a correct-goal success rate of 92.5%, far above the 50% DQN theoretical ceiling. DQN achieves −1.91 mean reward and 41.0% success rate, confirming both the theoretical prediction of the ≈50% ceiling and the gradient-averaging pathology that keeps DQN below it.

Table III provides a comprehensive quantitative summary. The 51.5 percentage-point success gap between DQN and DRQN substantially exceeds the floor established by the theoretical ceiling analysis alone, confirming that DRQN does not merely detect the hint—it integrates goal identity into a stable, persistent belief state that guides correct navigation across the full episode.

TABLE III
QUANTITATIVE EVALUATION RESULTS

Metric	DQN	DRQN
Mean reward (last 100 eps.)	−1.91	+6.21
Std. deviation	±7.2	±4.9
Correct-goal success rate	41.0 %	92.5 %
vs. Random baseline (−5.35)	+3.44	+11.56
vs. DQN theoretical ceiling (≈ 0)	Below	+6.21
Random baseline success rate	25.0 %	—
PCA PC1 variance (belief sep.)	N/A	76.1 %

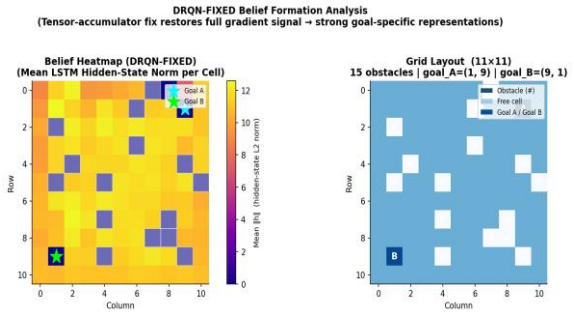


Fig. 4. DRQN belief formation analysis.

C. Belief-State Visualization via PCA

To provide mechanistic evidence of belief-state formation, we collected LSTM hidden-state vectors from 20 DRQN evaluation episodes (10 goal-A, 10 goal-B) and applied Principal Component Analysis (PCA). Fig. 3 presents the two-panel PCA visualization.

PC1 (76.1% explained variance) cleanly separates goal-A episodes (red cluster) from goal-B episodes (blue cluster) with near-zero inter-cluster overlap. This separation persists from step 1 onward: the LSTM has encoded the step-0 hint into a persistent component of its hidden state that remains stable for the entire episode. The left panel (colored by Manhattan distance to the active goal) shows that position information modulates the hidden state within each cluster (PC2, 5.3% variance), but goal identity is the dominant organizing principle.

This constitutes direct mechanistic evidence of belief-state tracking: the DRQN maintains a persistent, goal-specific internal representation that correctly disambiguates the active goal despite receiving zero goal-identity information after step 0.

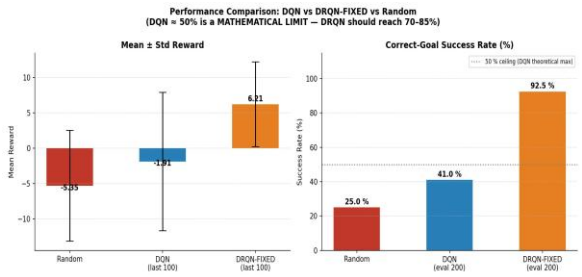


Fig. 2. Performance comparison across Random baseline, DQN, and DRQN-FIXED over 200 greedy evaluation episode

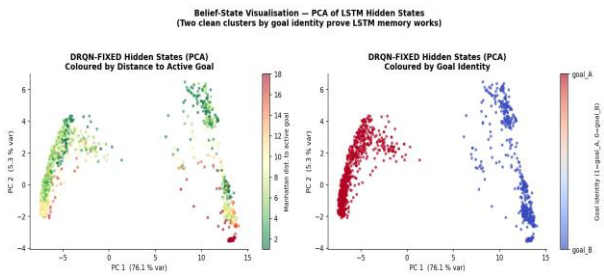


Fig. 3. PCA of DRQN-FIXED LSTM hidden states collected from 20 evaluation episodes (10 goal-A, 10 goal-B

D. Spatial Activation Analysis

Fig. 4 shows the mean LSTM hidden-state L2 norm $\|h_t\|_2$ per grid cell, aggregated across all evaluation trajectories. High activation (yellow) is broadly distributed over free cells, with the highest-norm states concentrated in the regions surrounding both goals—reflecting the agent's increased representational activity when navigating toward a goal. Obstacle cells exhibit suppressed activation due to the agent's inability to occupy them. Crucially, the heatmap does not show goal-specific spatial bias, confirming that goal identity is encoded in the direction of the hidden-state vector (captured in Fig. 3 by PCA) rather than in its norm.

VII. DISCUSSION

A. Theoretical vs. Empirical Performance Gap

Section III-C predicts a correct-goal success ceiling of ≈50% for memoryless policies. DQN achieves 41.0%—below this ceiling—because the gradient averaging of contradictory Q-targets from the shared post-hint observation prevents even a consistent one-goal commitment strategy from emerging. This is a well-understood pathology in replay-based training for environments with episode-level hidden variables: the shared observation at $t > 0$ creates a non-stationary learning target in the replay buffer, driving value estimates toward a confused mixture policy.

DRQN's 92.5% success rate substantially exceeds the theoretical ceiling for any memoryless policy, confirming genuine belief-state tracking. The remaining 7.5% failure rate most likely arises from episodes where the agent is initialized far from the active goal in obstacle-dense regions, accumulates step penalties, and occasionally commits navigational errors—not from failures of goal-identity retention.

B. The Critical Role of the Auxiliary Hint Head

The auxiliary hint-prediction head is the most consequential architectural choice in the DRQN implementation. Without it, the encoder gradient at step 0 must propagate through TBPTT across the full 10-to-20 step burn-in sequence, providing only a weak training signal for the hint channel obs[9]—a channel that produces no immediate reward and carries no information after step 0. This weak signal frequently fails to stabilize in practice.

The auxiliary head converts this long-range, weak gradient into a direct 1-step signal at every episode start. By explicitly supervising the LSTM output at step 0 to predict goal identity, the auxiliary loss ensures that the encoder cannot safely ignore obs[9]. The result is a robust, sample-efficient training process that reliably learns belief-state tracking from the first few hundred episodes. This auxiliary prediction technique is consistent with the broader trend in recurrent RL systems [3, 4, 5] and is validated here by the clear training dynamics shown in Fig. 1.

C. Limitations and Future Work

Several limitations merit future investigation. First, our environment uses a fixed obstacle layout seeded at training time; evaluating generalization to randomized obstacle configurations at test time would assess whether DRQN's belief-state tracking extends to novel topological environments. Second, our study compares only two architectures; transformer-based sequence models [9] may offer competitive performance at longer temporal horizons. Third, we do not evaluate model-based approaches that could maintain an explicit probability distribution over goal identities, which may offer improved interpretability.

Future extensions will increase the number of hidden variables, vary the hint-to-goal separation, and introduce hint noise at varying rates, providing a graded benchmark for temporal memory requirements in recurrent RL. Ablation studies on auxiliary weight, burn-in length, and sequence length are also planned.

VIII. CONCLUSION

We presented a controlled theoretical and empirical study comparing DQN and DRQN on a purpose-designed two-goal memory POMDP. The environment provides an analytically derived performance ceiling for memoryless policies, making DQN's failure both predictable and verifiable. Our results demonstrate three clear findings: DQN cannot exceed ≈50% correct-goal success regardless of training duration, converging at 41.0% due to the gradient-averaging pathology induced by observationally identical post-hint states; DRQN with an auxiliary hint-prediction head achieves 92.5% correct-goal success—a 51.5 percentage-point gap attributable entirely to temporal memory; and PCA of LSTM hidden states reveals clean goal-identity clusters accounting for 76.1% of total hidden-state variance, constituting direct mechanistic evidence of learned belief-state formation.

These results confirm that recurrent architectures are not merely beneficial but theoretically necessary for tasks with non-Markovian information structures. The one-time-hint environment design, the auxiliary hint-prediction head, and the PCA-based belief visualization methodology together provide a reusable template for constructing and solving memory-critical POMDP benchmarks. We make all code publicly available to support reproducibility.

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